



# FX PB Collateral

Automating FX Prime Brokerage Monitoring

Replacing tedious manual portal extractions with an automated end-to-end data pipeline & Power BI analytics dashboard



# The Business Problem & Enterprise Impact

Understanding the manual bottlenecks in daily FX Prime Brokerage collateral and limits reporting.



# Manual Process Challenge

1

## Portal Authentication

Operations team had to manually log into the Prime Brokerage (PB) portal every morning, navigate through reporting sub-menus, and await slow report loads.

2

## Manual Screenshotting

Gather relevant quantitative and qualitative data from internal and external sources

3

## Teams Pasting & Delays

Clean incomplete records, remove duplicates, and ensure accuracy to maintain data integrity



# Implemented Solution

Automated Ingestion Pipeline



## Automated Email Delivery

PB portal configured to auto-distribute daily CSV files



## Power BI Dataflows

Connects to SharePoint, cleans, aggregates, and structures the raw CSV data for downstream reporting



## Power Automate Workflow

Cloud workflow automatically saves CSV files to a secured SharePoint folder



## Power BI Reports

Automatically refreshed interactive reports distributed to the primary stakeholders

# Operational Speed & Impact

**<10s**

Time to Decision  
Insight

## **Instant Collateral Requirements Visibility**

By removing manual portal logins, screenshotting, and Teams messaging, the solution compresses the daily reporting cycle from over 15 minutes down to under 10 seconds.

Stakeholders receive updated and reliable collateral report every morning.

# Public Portfolio & GitHub Replication

Enterprise project securely recreated into an open-source demonstration





# Enterprise vs Open Source



| Workplace Production             |            | GitHub Replication                                |
|----------------------------------|------------|---------------------------------------------------|
| Prime Brokerage portal CSV feeds | Data       | Custom Python script generating synthetic FX data |
| Microsoft SharePoint Online      | Storage    | PostgreSQL on Neon Database                       |
| Power BI Dataflows to BI reports | Connection | Direct Power BI connection to Neon DB.            |
| Power BI Report                  | Reporting  | BI Tools – Power BI or Google Data Studio         |

# Data Architecture



Python & Neon PostgreSQL Pipeline



## Data Generation

Developed a Python script to synthesize mock data mimicking original reports



## Database Integration

Structured a relational schema and automated data loads directly into a Neon PostgreSQL serverless cloud database



## Power BI Reports

Connected Power BI Reports directly to Neon Database to model data, write DAX measures, and publish the dashboard





# AI-Assisted Development

1

## Rapid Scripting & Synthetic Logic

Leveraged Generative AI assistance to program complex Python generator scripts

2

## Database Schema Optimization

Leveraged AI prompts to help design PostgreSQL tables— schema layout, primary keys, and data types for Neon DB

3

## Documentation

Generated docstrings and a comprehensive GitHub README explaining setup instructions for reviewers

4

## Implementation

AI accelerated implementation while designing, testing, and validating the logic.



# Reporting Examples

Interactive Power BI Reports





# Collateral Dashboard



## Collateral Dashboard

Selected Date

10/Aug/2026

Initial Margin

-5.09M

Change: ▲ 588K

Variation Margin

-46.24K

Change: ▼ 41K

Collateral Required

-5.13M

Change: ▲ 547K

Collateral Posted

6.19M

Change: ▲ 1M

Excess / Deficit

1.06M

Change: ▲ 2M

Total Portfolio (In USD)

102.93M

Change: ▼ 11M

Open Positions

45

Change: ▼ 1

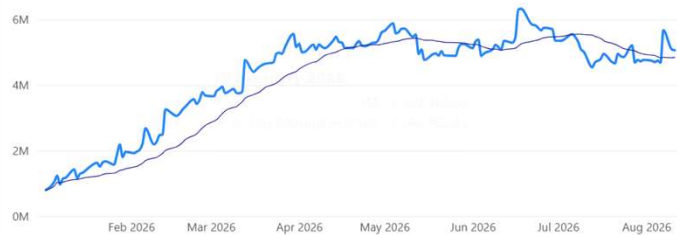
☐ Latest Date

☒ 10-Aug-2026

☐ 07-Aug-2026

### Initial Margin

IM vs 30 Day Moving Average



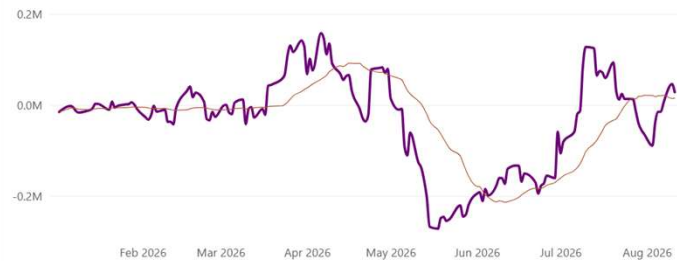
### Collateral: Required vs Posted

Posted vs Required



### Variation Margin

VM vs 30 Day Moving Average



### Excess / Deficit

